

Steward Jacob

Data Science & AI Student • Python • Machine Learning • Deep Learning • Power BI
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PROFESSIONAL SUMMARY

BCA (Data Science & AI) student at Yenepoya University with hands-on experience building ML models, deep learning systems, and AI-powered applications. Developed a CNN-based Pneumonia Detection web app achieving 89% accuracy and 97% pneumonia recall with Grad-CAM explainability; built a student performance regression model (Ridge R^2 0.88); and created an AI task assistant using Google Gemini API at a national hackathon. Proficient in Python, TensorFlow, Scikit-learn, and Power BI. Targeting data science and ML internship roles where analytical and AI development skills can deliver real-world impact.

TECHNICAL SKILLS

Languages: Python, R, SQL (basic)

ML / DL: Scikit-learn, TensorFlow, Keras, CNNs, Regression, Classification, Data Preprocessing, Model Evaluation

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Plotly, OpenCV, Pillow (PIL)

Visualization: Power BI, Plotly, Matplotlib, Seaborn, Microsoft Excel

Web / Deploy: Streamlit, Grad-CAM (XAI), Git, GitHub

IDEs / Env: Jupyter Notebook, Google Colab, VS Code

Concepts: Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, Basic CNNs, Explainable AI

EDUCATION

Bachelor of Computer Applications – Data Science & AI 2023 – 2026 (Expected)

Yenepoya (Deemed-to-be University), Bangalore | CGPA: 7.2 / 10

- Key Coursework: Machine Learning, Deep Learning, Big Data Analytics, Cloud Computing, DBMS, Python, Statistics for Data Science, AI
- Completed Data Science Workshop by industry professionals at Yenepoya University; produced live Power BI dashboard as deliverable

Higher Secondary Certificate (HSC) 2021 – 2022

Bishop Hodges Higher Secondary School, Mavelikara, Kerala | Percentage: 77.0%

- Subjects: Computer Science, Physics, Chemistry, Mathematics

Secondary School Leaving Certificate (SSLC) 2019 – 2020

St. John's Higher Secondary School, Mavelikara, Kerala | Percentage: 94.5%

- Subjects: Computer Science, Physics, Chemistry, Mathematics

PROJECTS

Pneumonia Detection System — Deep Learning Web Application

Tools: Python, TensorFlow, Keras, OpenCV, Streamlit, Grad-CAM, NumPy, Pillow

Built a CNN-based medical image classifier that detects pneumonia from chest X-ray images with 89% overall accuracy and 97% pneumonia recall, trained on the Kaggle Chest X-Ray dataset (5,216 images).

Implemented Grad-CAM (Gradient-weighted Class Activation Mapping) to generate visual heatmap overlays on lung regions influencing the model's prediction, adding clinical-grade explainability.

Deployed as a fully functional Streamlit web application; custom CNN architecture with tuned hyperparameters, input size 150×150 RGB.

GitHub: github.com/STXWARD/Pneumonia-detection

Student Performance Prediction — ML Regression Model

Tools: Python, Pandas, Scikit-learn, Matplotlib

- Developed a regression model predicting student math exam scores using demographic and academic features; applied data preprocessing, outlier removal, and feature selection on 1,000 real student records (Kaggle dataset).
- Evaluated Linear Regression, Ridge, and Random Forest algorithms; Ridge Regression achieved best performance with R^2 of 0.88 and RMSE of 5.32. Notebook includes feature importance charts and model comparison visualisations.
- All code documented, visualised, and version-controlled on GitHub with full README.
- GitHub: github.com/STXWARD/student-performance-prediction

Task Buddy AI — Conversational AI Task Manager

Tools: Python, Google Gemini API, Replit

- Built an AI-powered task management assistant enabling users to create, organise, and prioritise tasks entirely through natural language; integrated Google Gemini API for intent parsing and response generation.
- Presented at Vibehathon 2024 (Polaris School of Technology × Replit), a national-level hackathon; received official participation certificate.
- GitHub: github.com/STXWARD/Taskbuddy

Sales Performance Dashboard — Business Intelligence

Tools: Power BI, Microsoft Excel, DAX

- Built an interactive BI dashboard during a college Data Science Workshop, transforming raw sales data into actionable visual insights including revenue trends, regional breakdown, and top product categories.
- Implemented dynamic slicers, KPI cards, pie charts, and trend visuals; demonstrated end-to-end pipeline from raw data cleaning in Excel to final interactive Power BI presentation.

CERTIFICATIONS

Developing Security Automation Tools with Python for SOC Operations — TCS iON Digital Learning (AIP 135) *Mar2025*

Probability & Statistics for ML & Data Science — Coursera / [DeepLearning.AI](https://www.coursera.org/learn/deep-learning-ai) *Feb 2025*

Basic Statistics in Python (Correlations and T-Tests) — Coursera *Feb 2025*

Interactive Statistical Data Visualization 101 — Coursera *May 2025*

Covid-19 Death Medical Analysis & Visualization using Plotly — Coursera *Mar2025*

ACHIEVEMENTS & CO-CURRICULAR

- **Vibehathon 2024** (Polaris School of Technology × Replit) — Presented Task Buddy AI at national-level hackathon; received official participation certificate.
- **IBM ICEDAY** — Participated in IBM's Innovation, Collaboration & Entrepreneurship Day with a poster presentation, gaining exposure to industry-level innovation showcases.
- **TCS iON AIP 135** — Completed industry-recognised program on Security Automation with Python, submitted project report for evaluation.
- **Data Science Workshop** — Completed hands-on workshop by professional data scientists at Yenepoya University; delivered a live Power BI sales dashboard as final output.

LANGUAGES KNOWN

English (Professional) • **Malayalam** (Native) • **Hindi** (Professional) • **German** (Basic — A2 Level)